CS683: Advanced Computer Architecture-2024

L10-Virtual or Physical, kya karein kya na karein

https://www.cse.iitb.ac.in/~biswa/courses.html

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Phones (smart/non-smart) on silence plz, Thanks

Recap as camera was not working properly ☺

- Paging
- Page size
- Page table
- Page table walk
- TLB (cache, indexed by few bits from virtual address)
- TLB reach

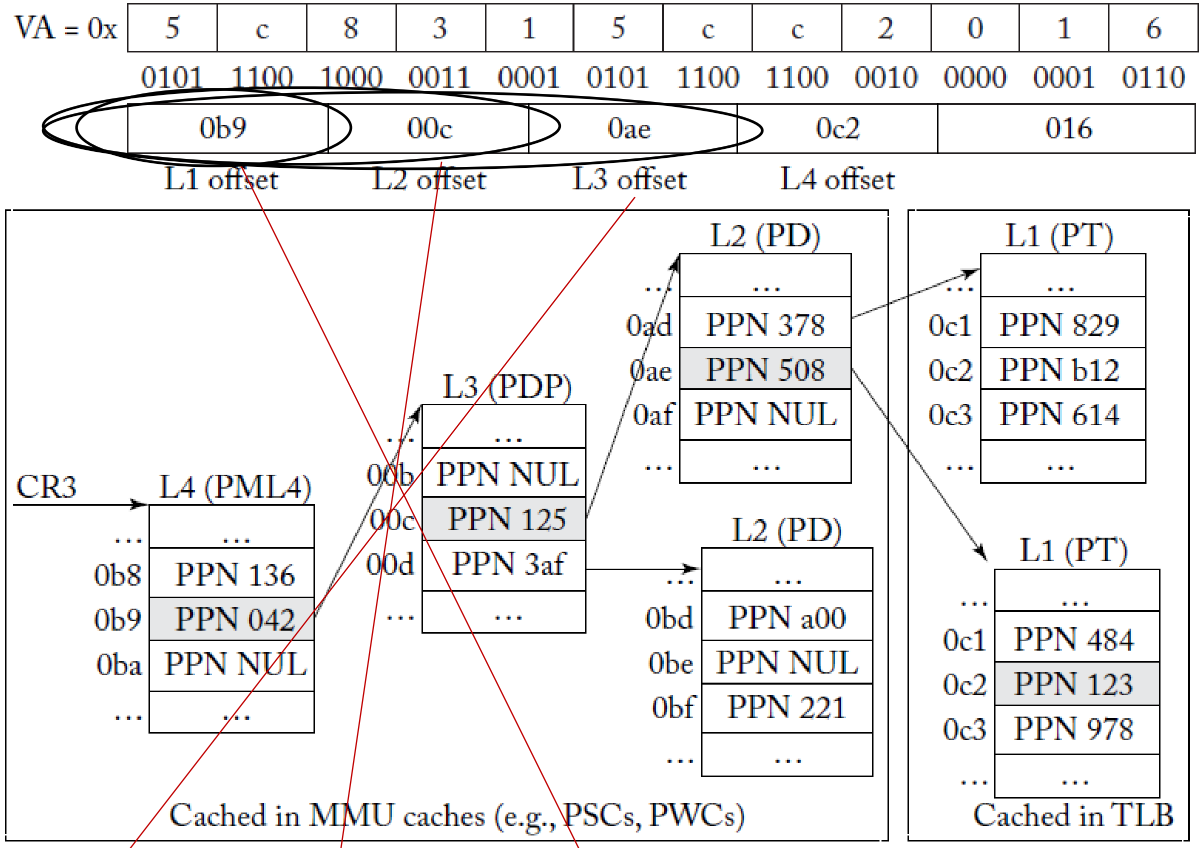
Some From ARM Neoverse

The data L1 TLB is a 48-entry fully associative TLB that is used by load and store operations. The cache entries have 4KB, 16KB, 64KB, 2MB, and 512MB granularity of Virtual Address (VA) to Physical Address (PA) mappings only. A hit in the data L1 TLB provides a single CLK cycle access to the translation, and returns the PA to the data cache for comparison. It also checks the access permissions to signal a Data Abort.

L2 TLB: Stores both data and instruction translations. 5-way, set associative, 1280-entry cache

https://en.wikichip.org/wiki/arm_holdings/microarchitectures/neoverse_n1#google_vignette

Small and subtle cache: MMU (PSC, PWC) Cache



PSC: L2

L4/L3/L2 Ind.	Page
b9/0c/ae	508
...	...

PSC: L3

L4/L3 Ind.	Page
b9/0c	125
...	...

PSC: L4

L4 Ind.	Page
b9	042
...	...

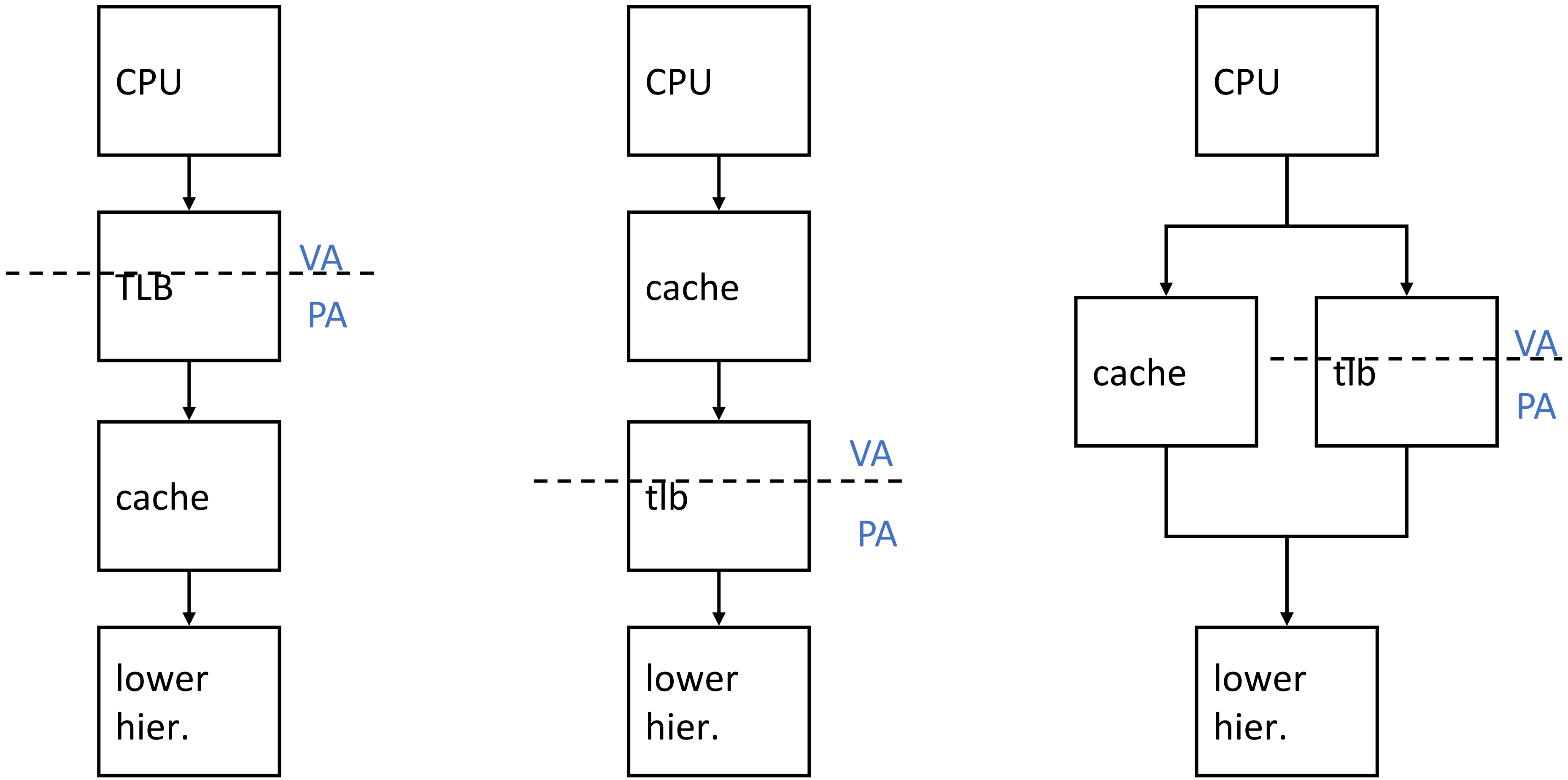
TLB

Virtual Page	Physical Page
b9/0c/ae/c2	123
...	...

More details

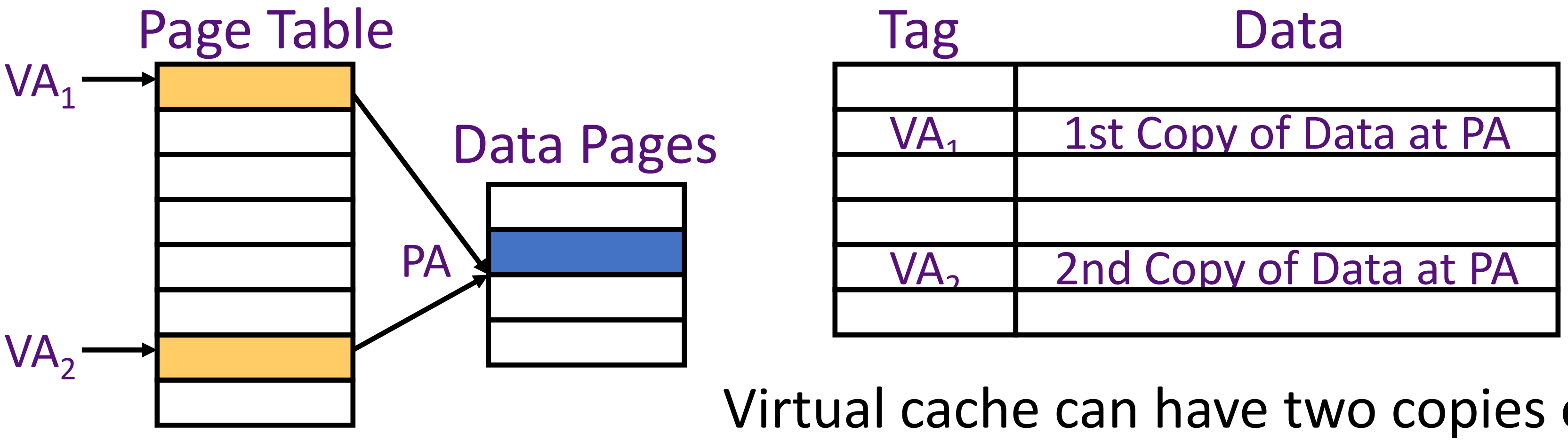
- MMU caches are searched parallelly.
- Anyways, these are small caches with few entries: 2, 4, 8, 16 entries kinda
- PSCL2 will have more entries than PSCL3, and PSCL4
- PSCL4 cache usually have two entries only ☺
- For five levels of PTW, we will have PSCL5, 4, 3, 2
- Finally, modern processors use multiple PTWs to provide address translation parallelism.

Caches: Virtual or Physical



physical cache | virtual (L1) cache | virtual-physical cache

Synonym Problem



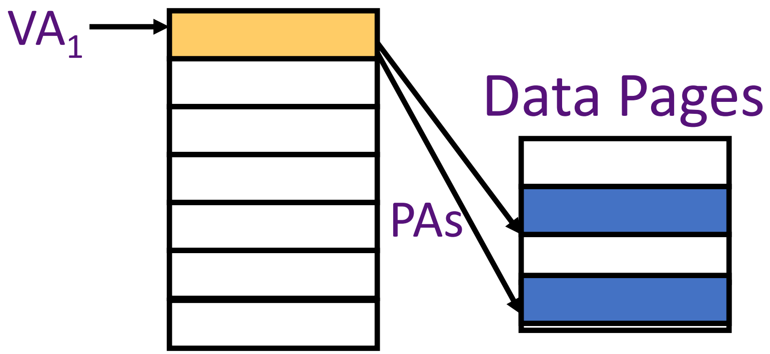
Two virtual pages share one physical page

Virtual cache can have two copies of same physical data. Writes to one copy not visible to reads of other!

General Solution: *Prevent aliases coexisting in cache*

direct-mapped cache can take care of it

Homonym Problem

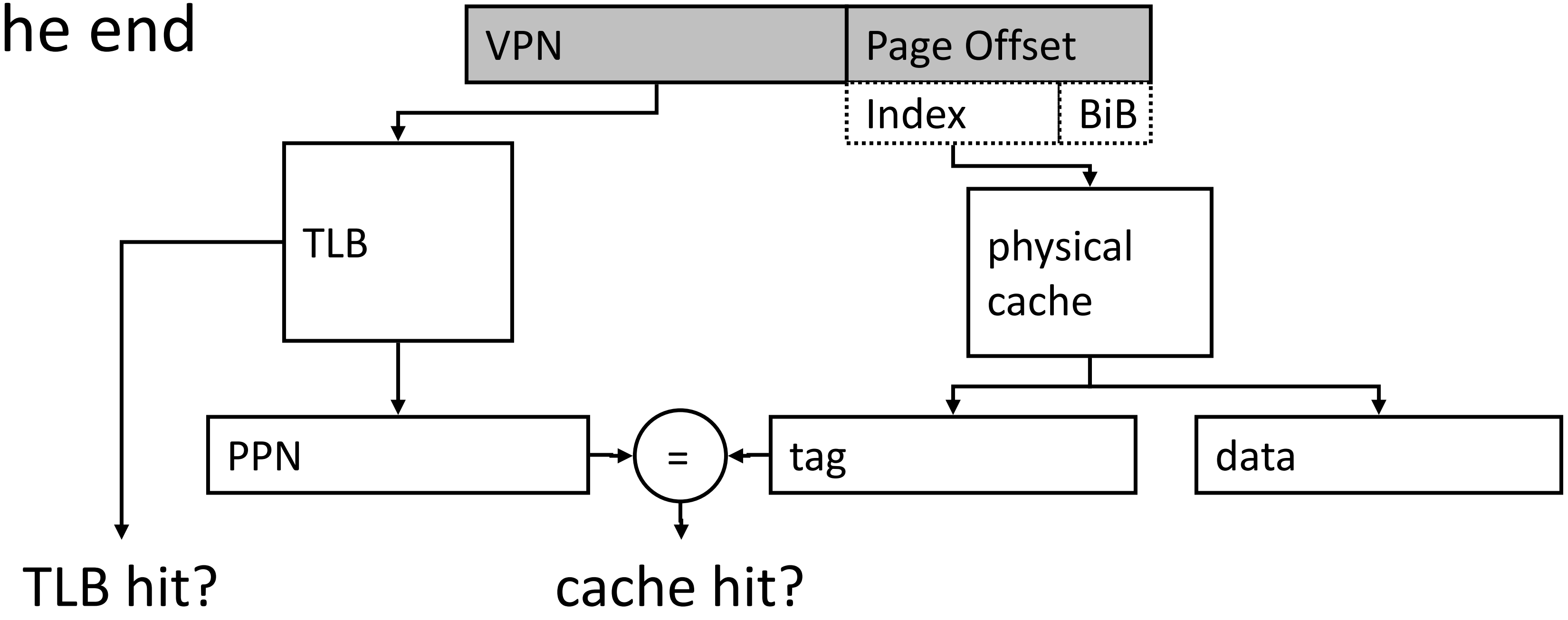


Tag may not uniquely identify cache data
Solution: Add ASID with tag
Or
Physical tags
Or
Flush on context switch

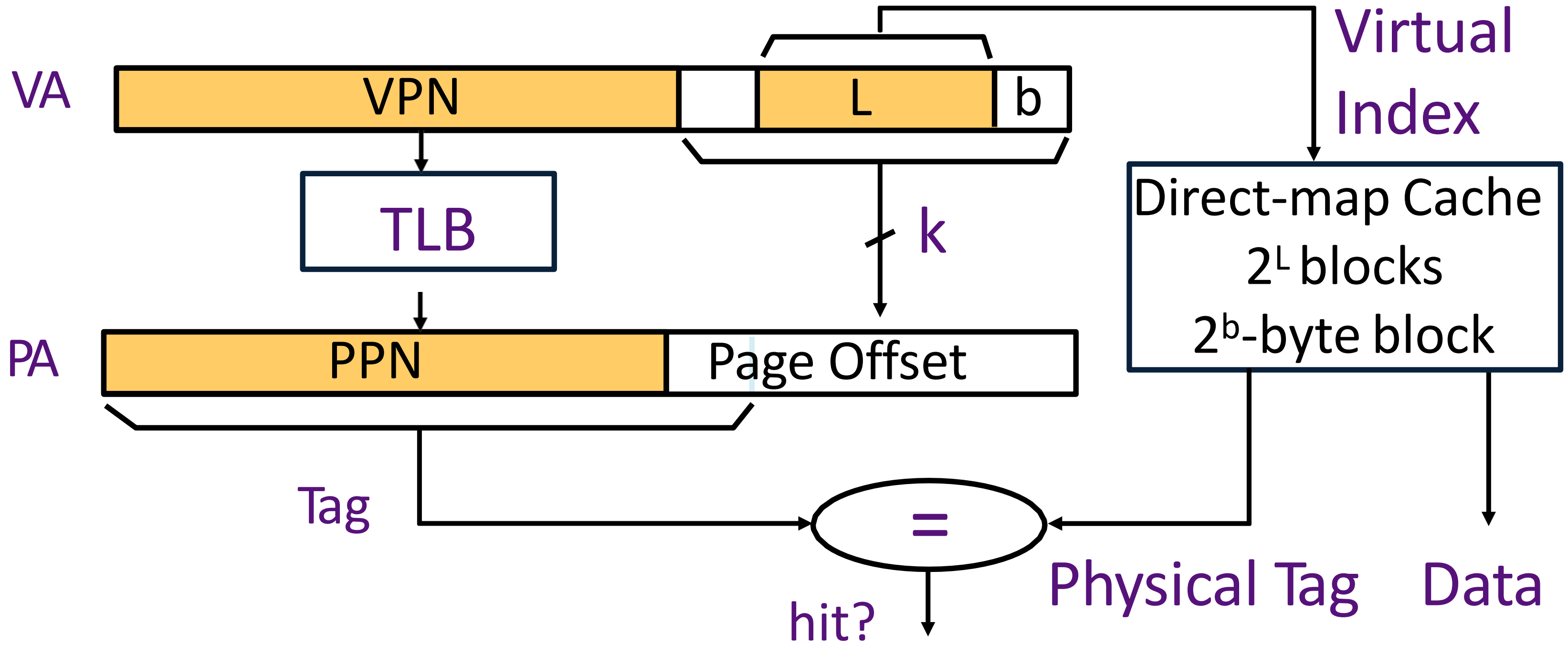
One virtual page maps to two physical pages

VIPT Caches

- If Cache size ≤(page_size × associativity), the cache index bits come only from page offset (same in VA and PA)
- If both cache and TLB are on chip: index both arrays concurrently using VA bits, check cache tag (physical) against TLB output at the end



Concurrent Access to TLB & Cache (Virtual Index/Physical Tag)



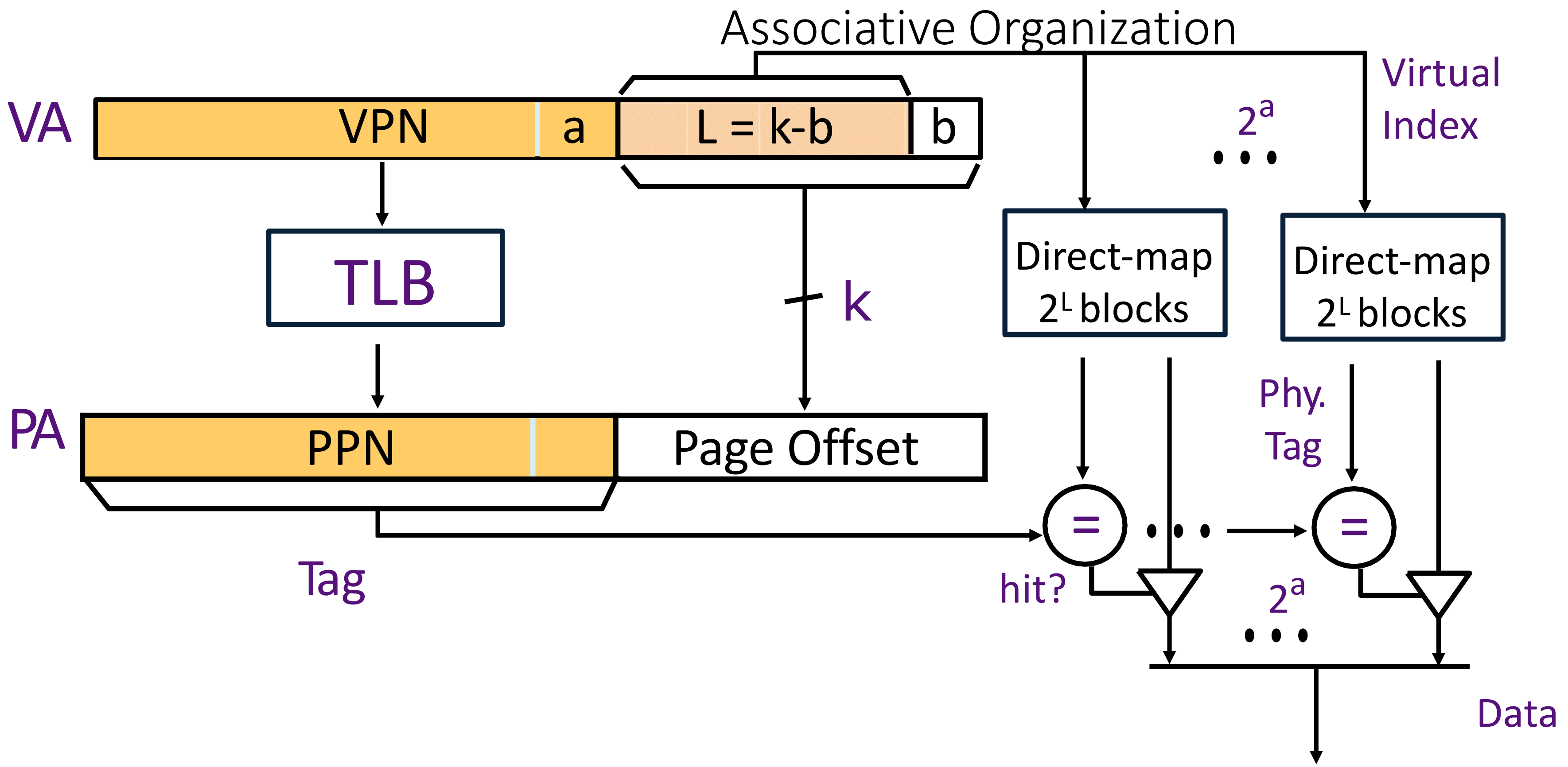
Index L is available without consulting the TLB

→ *cache and TLB accesses can begin simultaneously!*

Tag comparison is made after both accesses are completed

Cases: $L + b = k$, $L + b < k$, $L + b > k$

Virtual-Index Physical-Tag Caches:

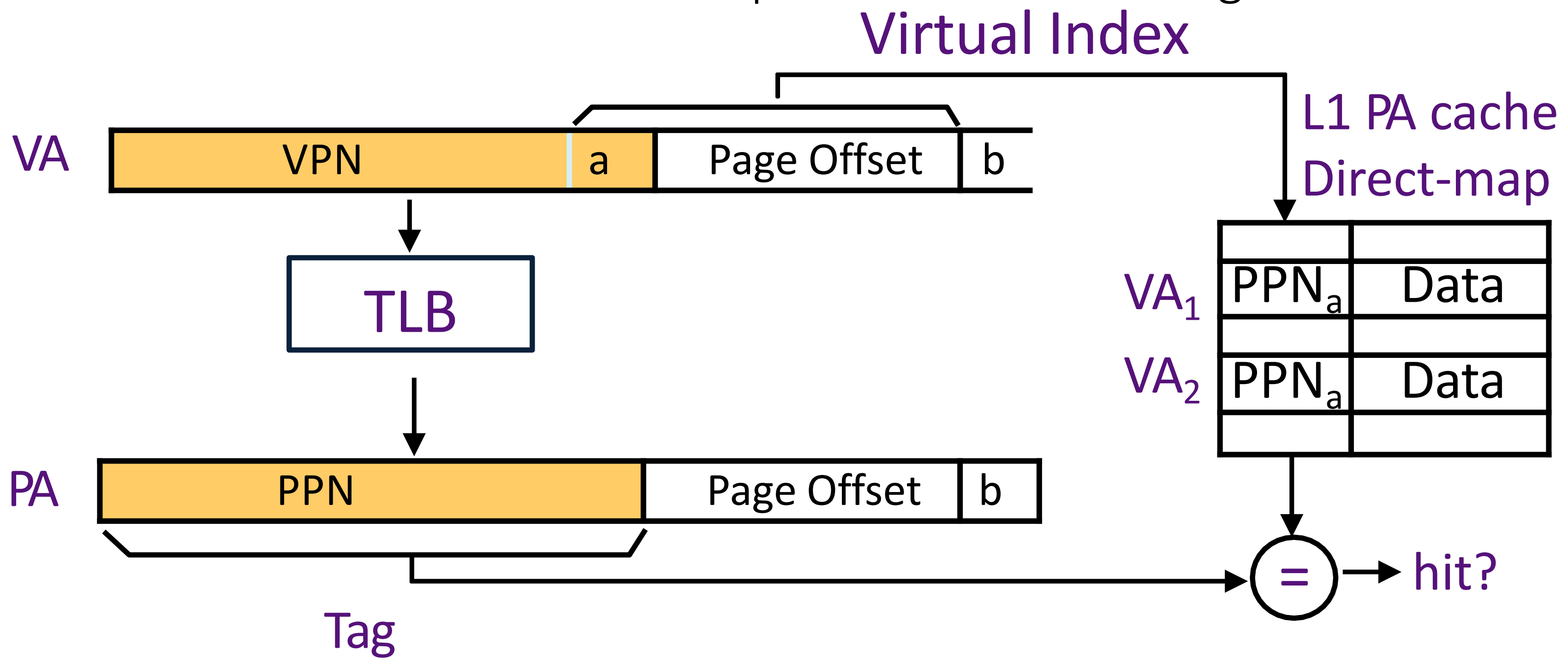


After the PPN is known, 2^a physical tags are compared

How does this scheme scale to larger caches? Think about it

Concurrent Access to TLB & Large L1

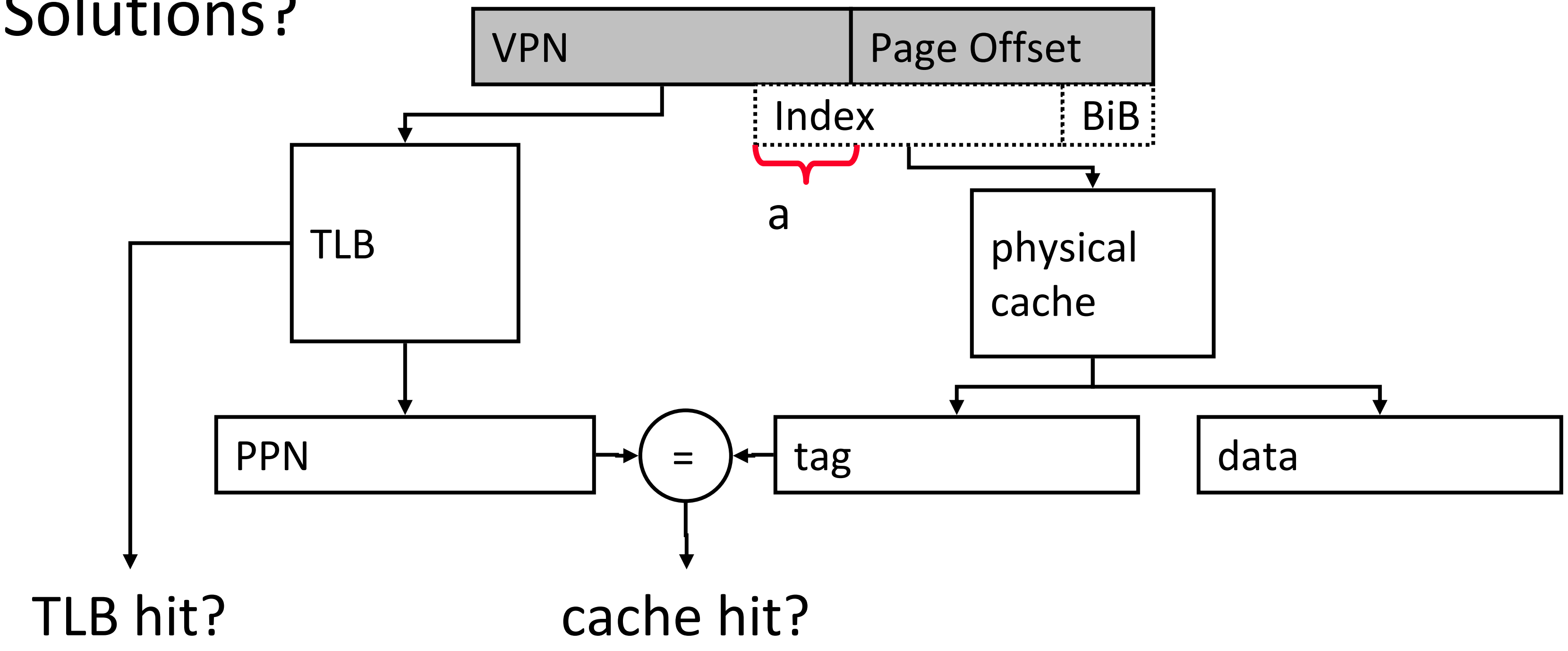
The problem with L1 > Page size



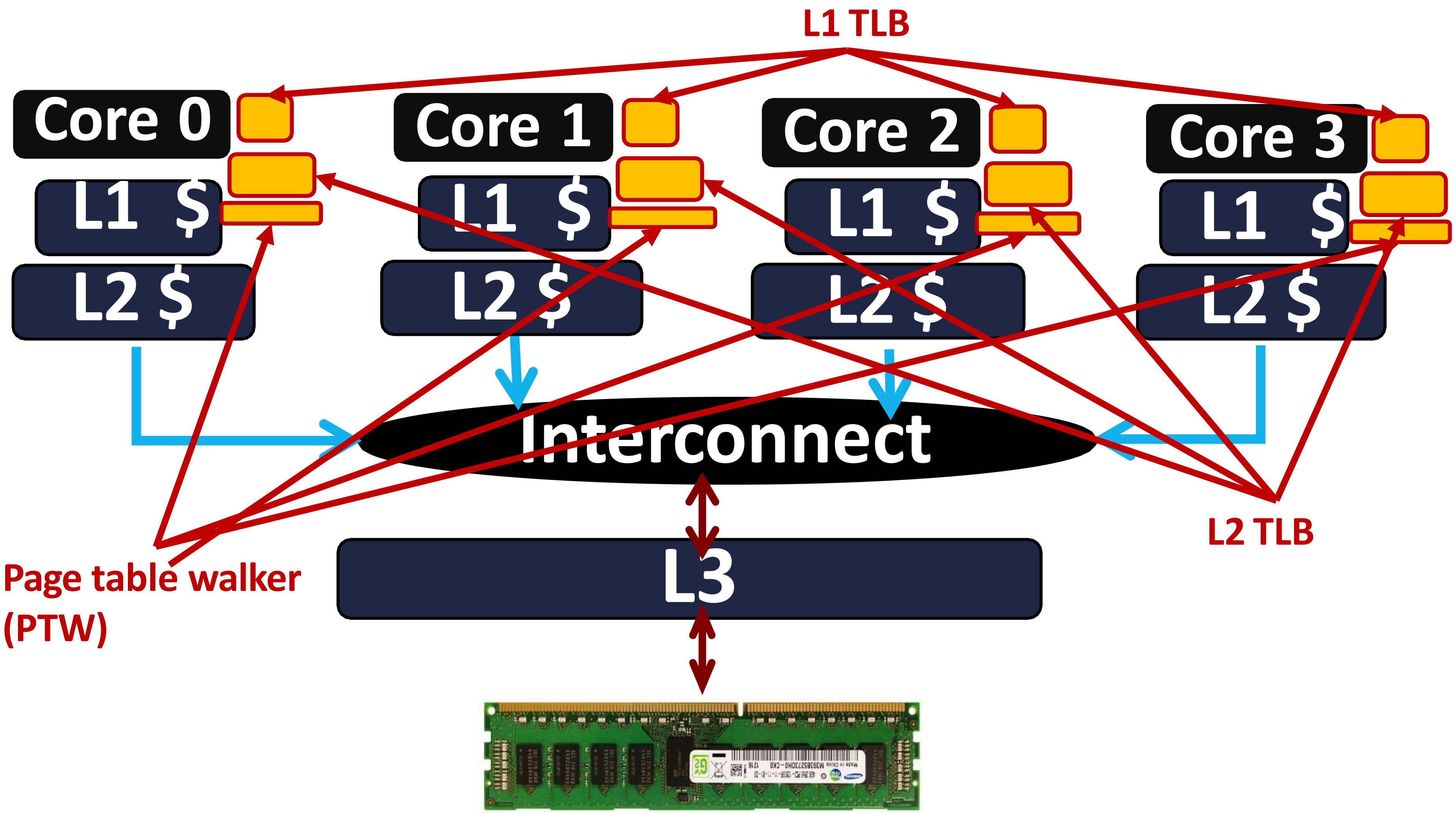
Can VA_1 and VA_2 both map to PA ?

What if? Think about PIPT, VIPT, PIVT, and VIVT

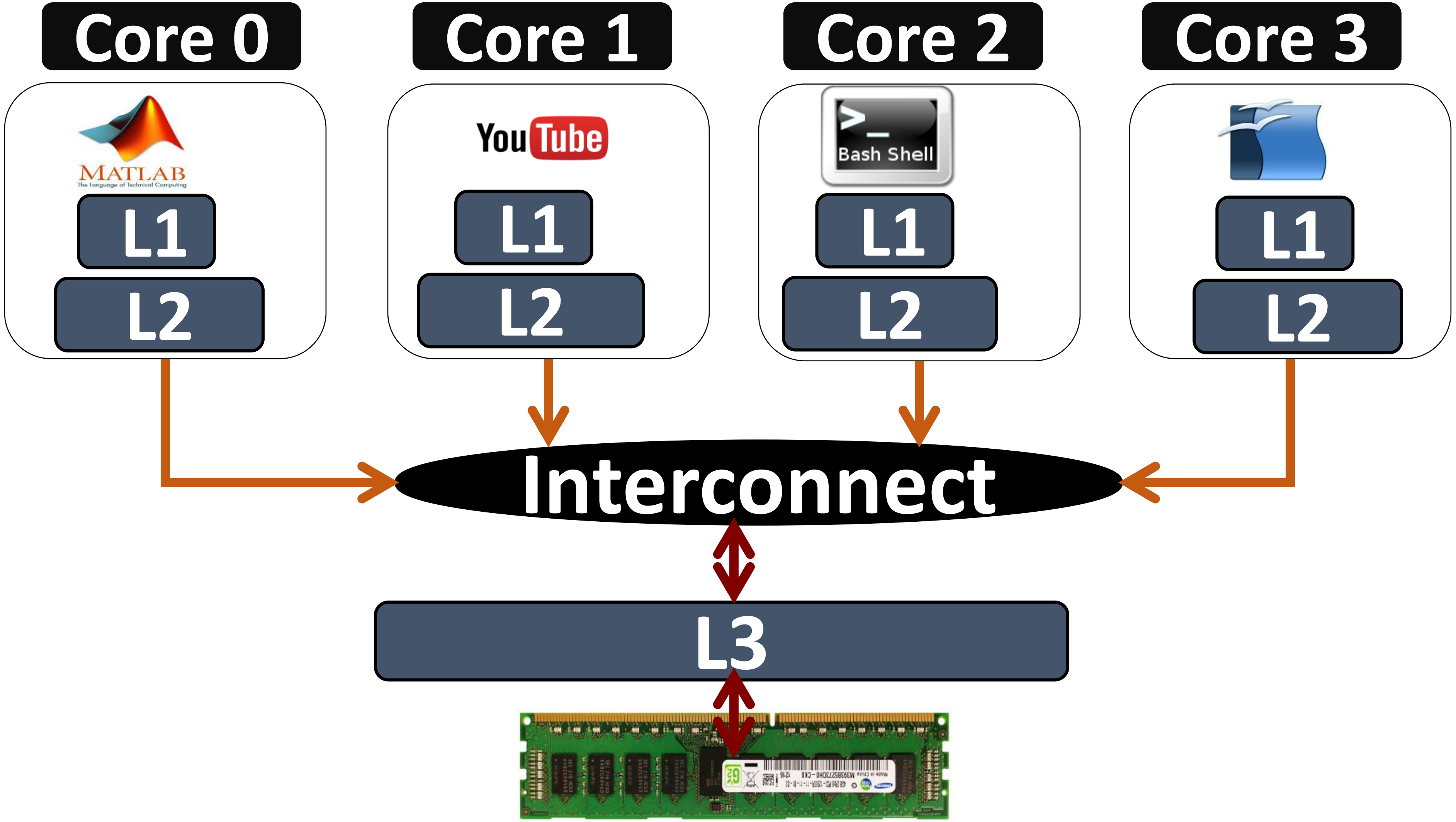
- If C>(page_size × associativity), the cache index bits include VPN ⇒ Synonyms can cause problems
    - The same physical address can exist in two locations
- Solutions?



Welcome to Multicore Systems



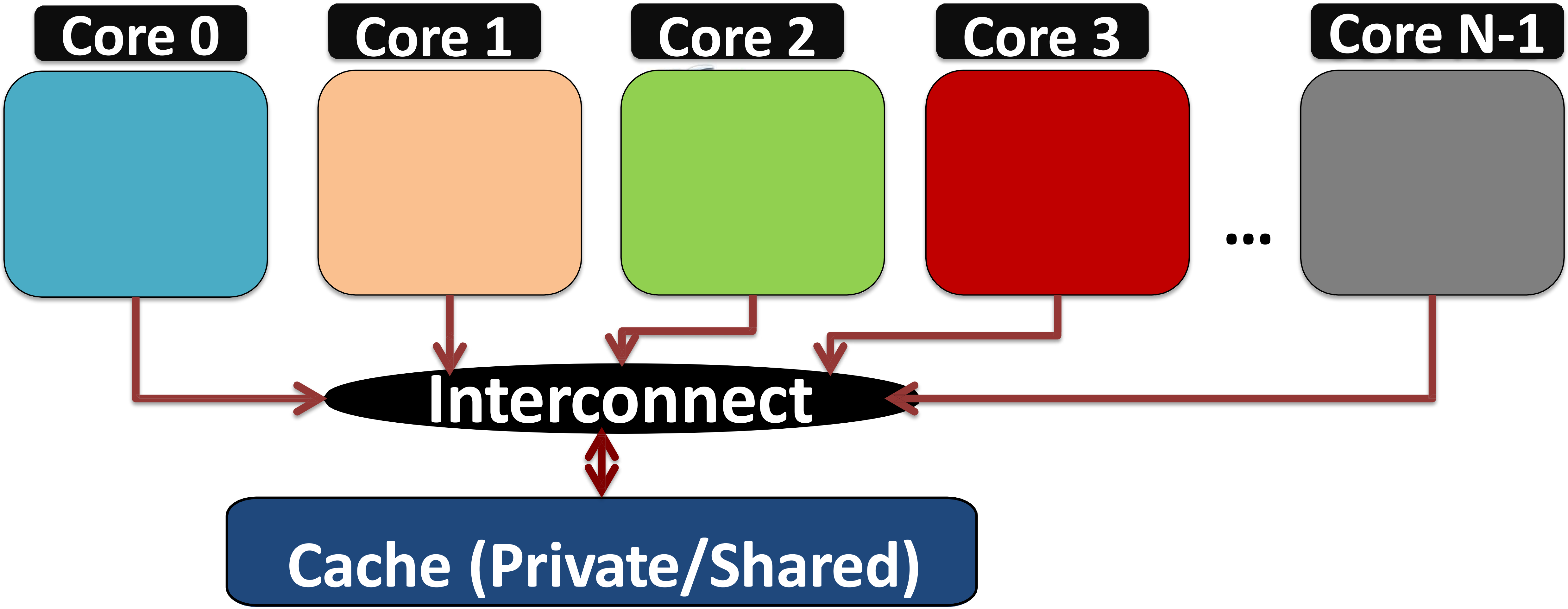
Multicore


Core 0
Core 1
Core 2
Core 3
MATLAB
YouTube
Bash Shell
L1
L2
L1
L2
L1
L2
L1
L2
Interconnect
L3

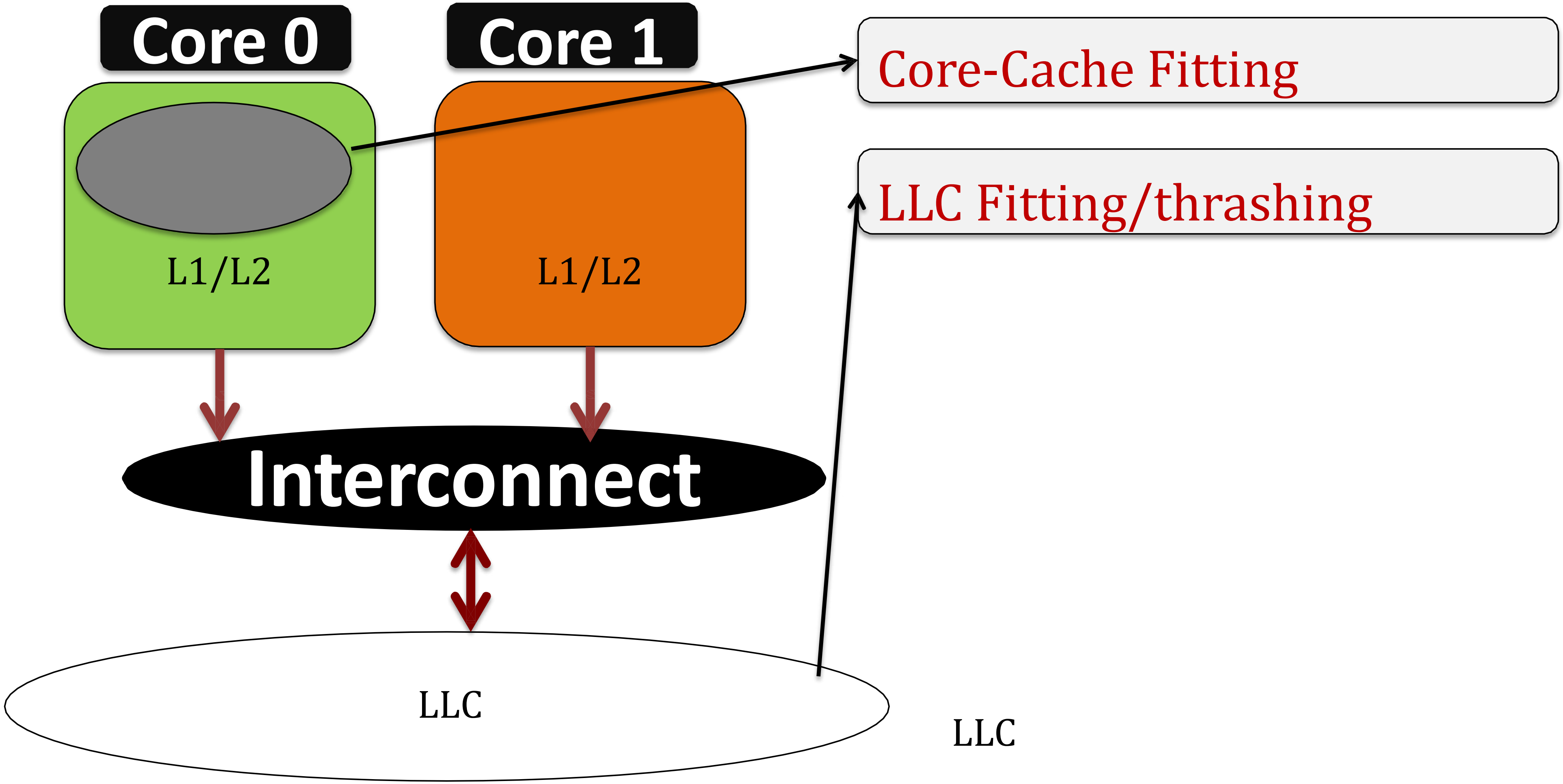
Oh hang on!! One more time, last time

- Do caches cache page table entries?
- Yes, there is no other way to go to memory
- So, the path for page table walk is: L1 TLB, L2 TLB, MMU caches, L1 cache, L2 cache, L3 cache, and then DRAM
- So the worst case of five memory accesses for one translation can be five L1 hits too ☺

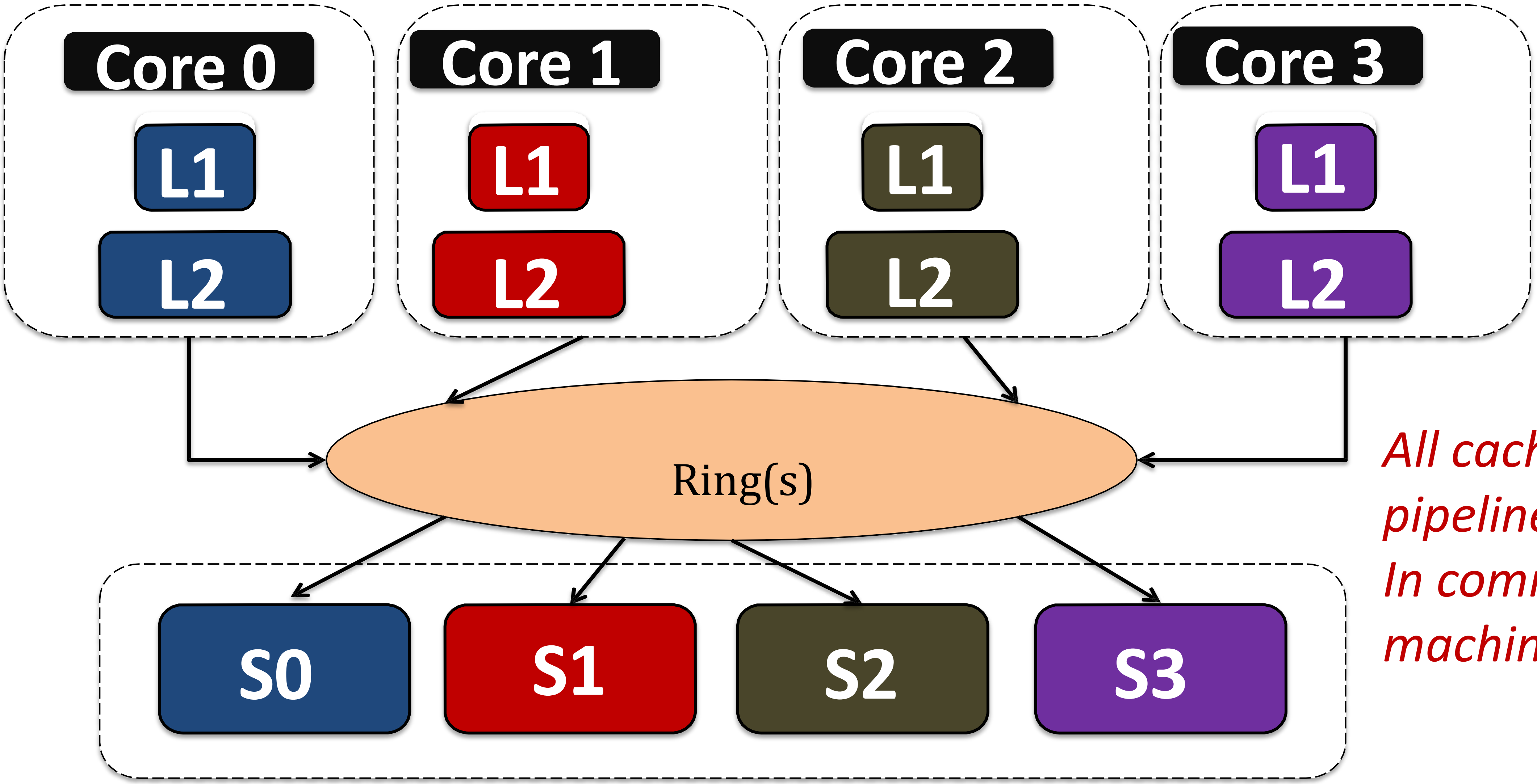
Caches: Private/Shared



Application behavior



Sliced/Banked LLC: NUCA (Non-uniform Cache)

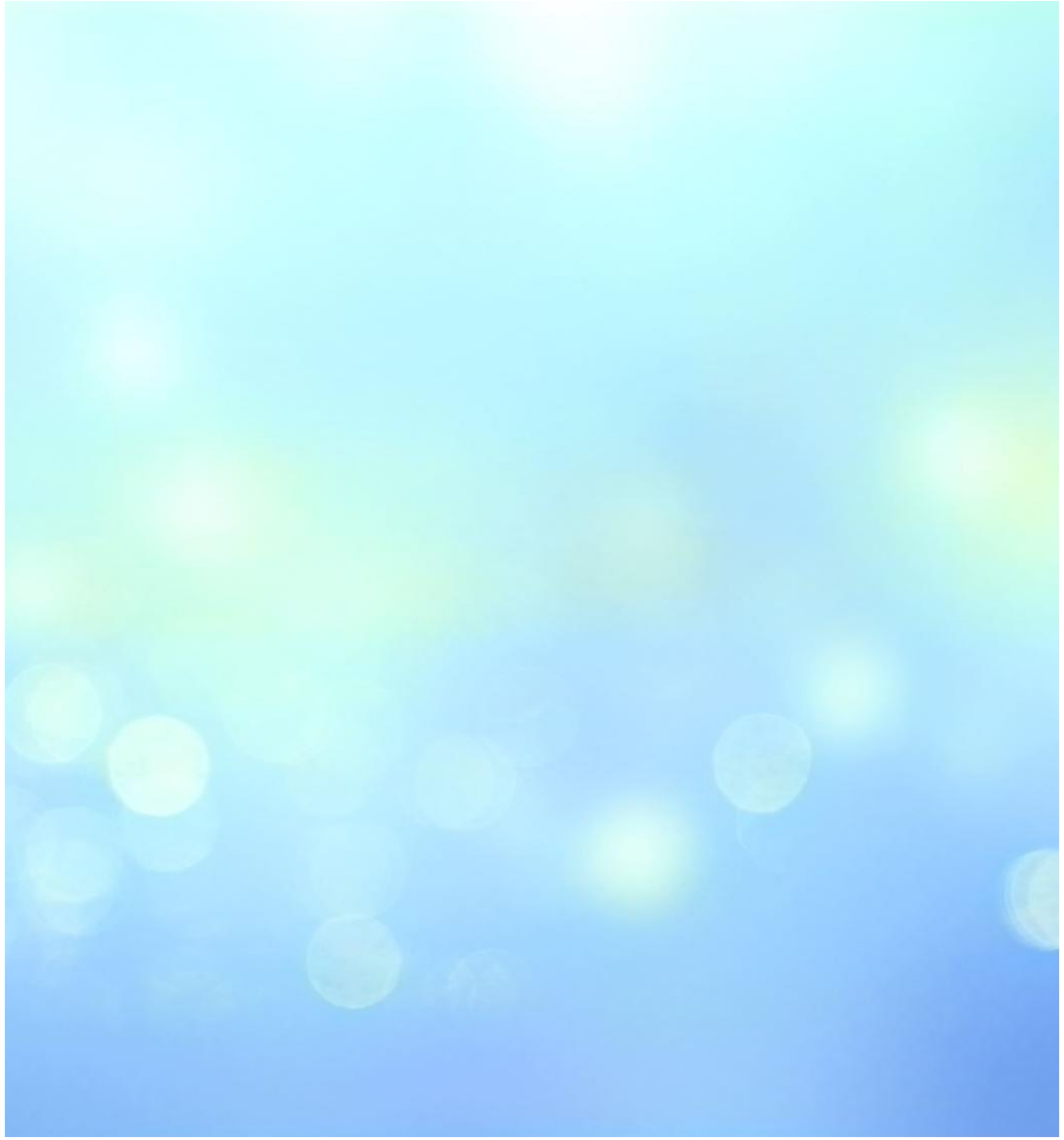


All caches are pipelined

In commercial machines

Shared Resources

- LLC
- Space contention
- Bandwidth contention (off-chip)
- Revisit replacement policies, prefetching etc



More Problems once we jump into DRAMs

Summary

- L1I
- L1D
- L2, LLC
- DTLB, STLB, L1, L2, LLC
- TLBs, MMU caches, Caches
- PTWs, TLBs, MMU caches, Caches
- VIPT, PIPT

Let’s Pause and Reflect on all

- Keywords: Latency, bandwidth, multi-level caches, TLBs, inclusive, exclusive, non-inclusive, inclusive, banking, pipelined, victim cache, stream buffers, prefetchers: software and hardware, stride, stream, next-line, SIMD, virtual memory, VIPT, PIPT, page table walkers, MMU caches, private cache, shared cache, sliced caches, streaming, thrashing, fitting, friendly, replacement policies: SRRIP/DRRIP, dead-block predictor, Miss rate, MPKI, MLP, MSHRs, fill buffers, Huhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhh